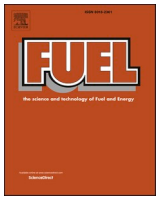
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Alternative fuels to reduce greenhouse gas emissions from marine transport and promote UN sustainable development goals

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ABSTRACT

The International Maritime Organization (IMO) has placed stricter controls on several aspects of global maritime transport operations to protect the environment. In light of this, the goal of this study is to examine and assess the different prospective paths and technologies that will assist the shipping industry in decarbonizing its operations. We consider how the utilisation of various alternative energy sources reduces greenhouse gas (GHG) emissions from marine transportation and contributes to the promotion of the United Nations Sustainable Development Goals (SGDs). The complexities associated with maritime industry operations using alternative energy sources are also explored. Biofuel as an alternative energy source, including biomethanol and biodiesel, can reduce greenhouse gas emissions in the shipping industry by 25% to 100%. However, the current supply of biofuels can only meet about 15% of the total demand which is not sufficient to sustainably power the entire marine fleet. There are several issues associated with these biofuels, including oxidation, ecological consequences, feedstock availability, technical and operational constraints, and economic factors that must be addressed before their full potential may be achieved.

1. Introduction

The expansion of both economies and populations are key contributors to the rising demand for energy around the world. They have been responsible for a large expansion of worldwide seaborne trade as well as a rise in the total number of boats engaged in global shipping [33]. Shipping is responsible for 80–90 % of global trade, which involves the annual movement of about 10 billion tonnes of containers carrying both solid and liquid bulk cargo across the world's oceans [28,91]. International trade grew by 18 % between 2016 and 2019, a trend resulting in a 50 % increase in the use of shipping fuel between 2012 and 2040 [64]. This industry is dominated by the use of fossil fuels with a high sulphur content, specifically heavy fuel oil. The emissions produced by a ship burning fuel with a sulphur level of 3.5 % are similar to those of 210,000 trucks [80]. As a result, shipping's contribution to global anthropogenic emissions has climbed to 2.89 %. Greenhouse gas emissions produced by ships, including sulphur oxides, nitrogen oxides, and particulate matter, pose a grave threat to the global environment and human health within 400 km of land [5].

According to the third greenhouse gas study conducted by the IMO in 2014, global commerce growth may result in a 50 % to 250 % rise in shipping emissions by 2050 [83]. In the same period, they anticipate

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that around 15 % of worldwide carbon dioxide emissions will relate to shipping. Shell and Deloitte [103] conducted a market study to gain a better understanding of the current industry trends. When asked about the importance of decarbonization for their companies, >90 % of those working in the shipping industry responded that it was either important or the top priority. Additionally, 80 % of respondents mentioned that the significance of decarbonization has expanded substantially over the course of the previous eighteen months [103]. This demonstrates that the market and industry are taking decarbonization into account as part of their long-term business plans, in keeping with the IMO's goal to cut carbon dioxide emissions from shipping by at least 50 % by 2050 compared to a 2008 baseline [39].

A significant reduction in carbon emissions calls for the establishment of new financial incentives and policies at regional, national, and worldwide levels [10,67]. It is usual practise to classify maritime emission and reduction strategies into the two primary categories of technical and operational [13,76]. The International Transport Forum recognises that there is more than one way to reduce carbon emissions, including the use of alternative fuels (such as sustainable biofuels, hydrogen, and ammonia), electrifying ships, and making use of wind power, though these could also be considered technical measures [24,47]. Fig. 1 shows a synopsis the short, medium, and long term steps to achieve an initial strategy to reduce greenhouse gas emissions from maritime transport [56].

To provide a smooth and dependable service in the most effective manner, the maritime transportation system must provide the safe, secure, efficient, and dependable transit of commodities throughout the globe, while limiting pollution, maximising energy efficiency, and assuring resource conservation [104]. To do this, the intricacy of the interrelationship between maritime transportation system participants must be acknowledged and accounted for when addressing specific acts. A sustainable maritime transportation system requires well-organized administrations that collaborate worldwide and promote adherence to global norms, with the assistance of institutions with relevant technical skills. To achieve rigors pollution control regulations, innovation and new technologies are required, such as on-board treatment systems and new engine technology. The quality of fuel oil has a direct correlation with the quality of its emissions; hence, appropriate quality standards for fuel oil should be imposed [105].

Recent studies estimate that the use of alternative marine fuels has the potential to reduce CO_2 emissions by 20–100 %, with the precise value depending on the fuel type [23,96]. Marine fuels like heavy fuel oil, liquefied natural gas, fossil methanol, biofuels, and liquefied hydrogen were analysed in a carbon-constrained environment using a global energy model [86]. The results showed that the greatest cost savings would be achieved by making the transition to liquefied natural gas and methanol as soon as possible, while using biofuels in the shipping industry was found to be inappropriate due to their low competitiveness and limited availability.

Several studies have analysed potential marine fuel alternatives based on various decision-making strategies. Conventional practise has relied on qualitative evaluations being informed by existing literature and practical experience in the field [15,44,96]. However, no definitive alternative marine fuels for future shipping carbon reduction plans were identified. Therefore, the purpose of this brief analysis is to shed light on the ways in which alternative energy sources can play an important role in reducing the amount of carbon dioxide emissions produced by the shipping industry as well as the potential obstacles that may arise. In this study, we identify the areas with the highest potential for emissions reduction as a guide towards further developing low-carbon shipping.

2. Maritime transport and associated emissions

The movement of nearly 10 billion tonnes of solid, and liquid bulk goods across the world's oceans annually is driven by maritime transportation, which accounts for 80–-90 % of worldwide trade [12]. The sea has been used throughout history as a mode of transporting people and goods across oceans and between continents. Although maritime transportation is capable of effectively and efficiently moving practically any commodity, the introduction of air travel has proven to be superior for transporting goods that are either perishable or of a high value. Providing electricity for propulsion, auxiliary, and ancillary applications in maritime transport can be accomplished through the utilisation of a variety of different technologies. One of the sources of power that ships draw from is the internal combustion engine (ICE). Nearly all of the world's merchant fleet is currently operated using marine ICEs, which gives them a dominant market share [18]. Also, at this time, the internal combustion engine remains the most effective mode of transportation in the shipping industry. The functioning of the ICE is straightforward, sophisticated, and reliable, and has a long lifetime sometimes comparable to the lifetime of a ship [18]. At the moment, there are many different kinds of internal combustion engine available.

Ship propulsion can also be accomplished using gas turbines, which

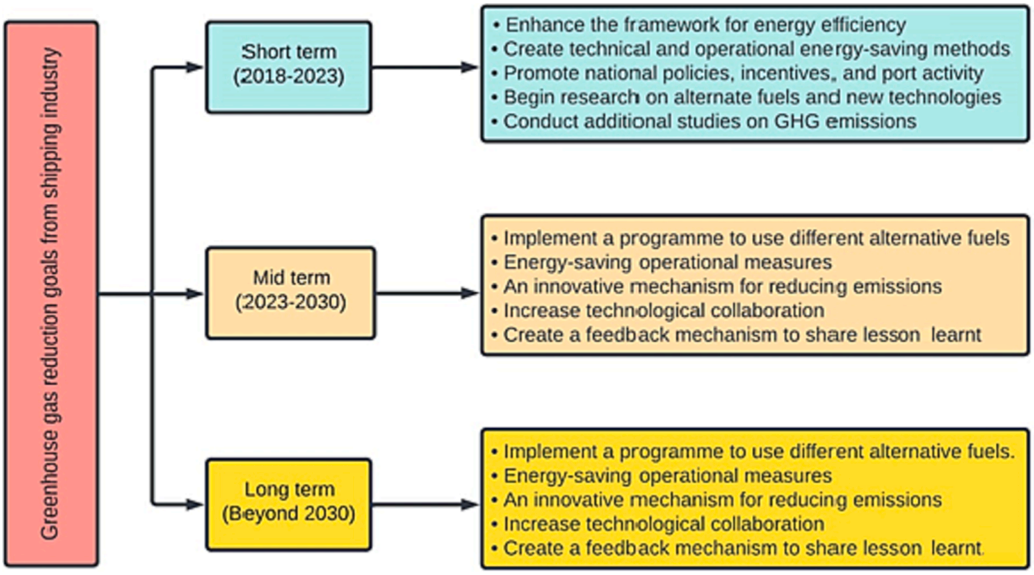


Fig. 1. Synopsis of short, medium, and long term steps to achieve an initial strategy to reduce greenhouse gas (GHG) emissions from maritime transport [56].

have found a home in the military sector due to the relatively short time they require to get started and their capacity to achieve high speeds [7]. However, gas turbines have a poor efficiency rating in environmental terms. A ship can also be powered using another type of technology known as fuel cells. Fuel cells are a proven technology for land-based power generation that could one day replace internal combustion engines. Fuel cells turn fuel into energy. Even though electricity is utilised for auxiliary purposes on ships, recent developments have demonstrated that it may also be used for propulsion [90,101]. Nuclear generators can also be used to power ships. Scientists have been experimenting since 1995 on ways to propel ships using nuclear power, mostly for use in naval ships and submarines [20]. Nuclear propulsion is accomplished by an onboard nuclear power plant that heats steam, which in turn powers steam turbines and generators. Nuclear propulsion is appealing for a number of reasons, including its high power density, low and steady pricing, and low greenhouse gas emissions [68].

Maritime transportation is typically seen as having a lower impact on the environment than other modes of travel, such as air and road transport. In spite of the relatively high efficiency of current propulsion systems, the use of heavy fuel oil, which is regarded to be a low quality grade fuel, results in substantial pollutant emissions and subsequently highly negative environmental consequences [88]. There has been a steady increase in both globalisation and the volume of international trade since the beginning of the industrial revolution which has resulted in an increase in the amount of emissions produced by maritime shipping [39]. Maritime transportation is responsible for between 14 and 31 % of the world's nitrogen oxide emissions and between 4 and 9 % of sulphur oxide emissions [44]. The marine industry uses 330 Mt of fuel annually, with heavy fuel oil making up the majority (77 %) of this total [82]. It is believed that this energy demand is responsible for 2–6 % of global carbon dioxide emissions [44]. Assuming business continues as usual, it is anticipated that carbon dioxide emissions from shipping would increase to a range of 1.1–3.7 Gt CO_2/year by 2050; this represents a 270 % increase in comparison to 2007 [79]. The majority of shipping-derived greenhouse gases can be attributed to bulk carriers, oil tankers, and container ships (Fig. 2), with faster ships producing less carbon dioxide than slower ones.

3. Environmental and health impact of shipping emissions

The shipping industry is responsible for a wide variety of detrimental effects on the marine environment (Fig. 3), some of which include the release of toxins from untreated wastewater, bilge and scrubber water, sewage, and antifouling paint [35–36]. Scrubber wash water and SO_X deposits contribute to acidification, hull fouling and ballast water can spread invasive species, and noise pollution and bilge water can disrupt marine ecosystems [58,78,87,100]. Additionally, shipping has an effect on natural systems through the processes of eutrophication and acidification [54,55].

The production of air pollutants like non-methane volatile organic compounds, PM, NO_X, and SO_X has an effect on human health, and shipping is one of the industries that contributes to this problem (Fig. 4) [9]. The emission of greenhouse gases like CO_2 into the atmosphere and short-lived climate pollutants like methane and black carbon, NO_X, and SO_X in diverse maritime locations are all factors that contribute to climate change [37,43]. According to the results of a study conducted by the World Health Organization, shipping was found to be the third most significant cause of air pollution [25]. This has occurred even after the development of the heating and air conditioning industry as well as the road transportation industry. This presents a novel potential threat to the health of those residing in coastal urban centres. Increases in the frequency and intensity of asthma attacks and the incidence of lung cancer are indicators of this threat. It also causes a surge in the incidence of cardiovascular disease, hospitalisation rates, and unexpected deaths

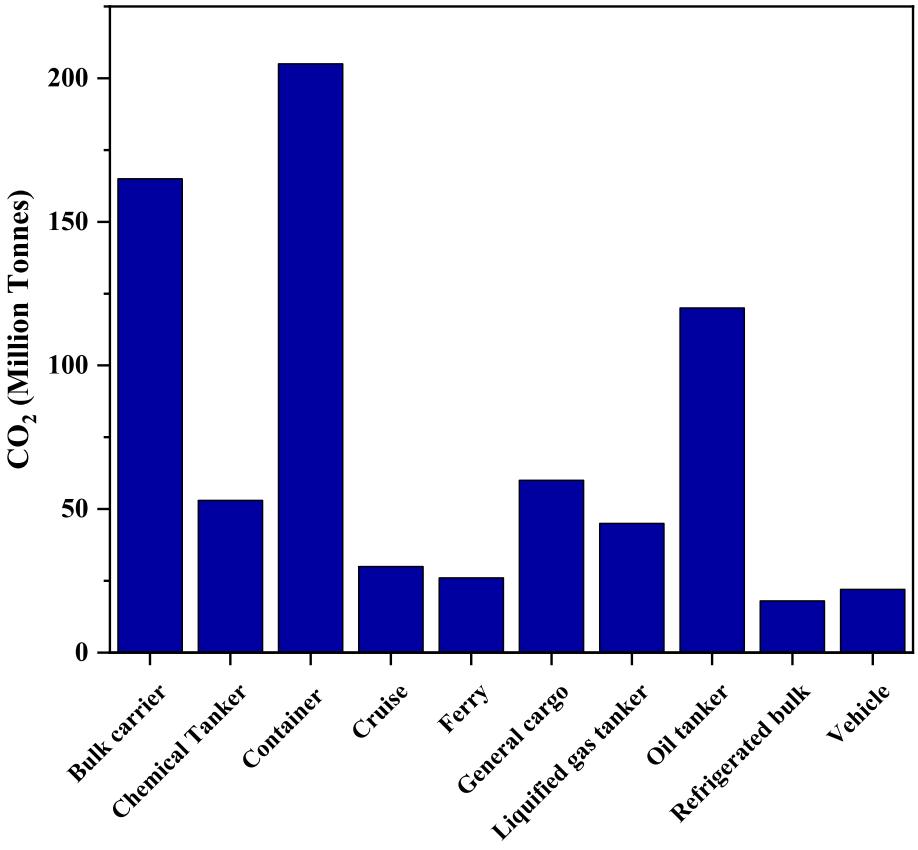


Fig. 2. Global CO_2 emissions from various maritime transport types [85].

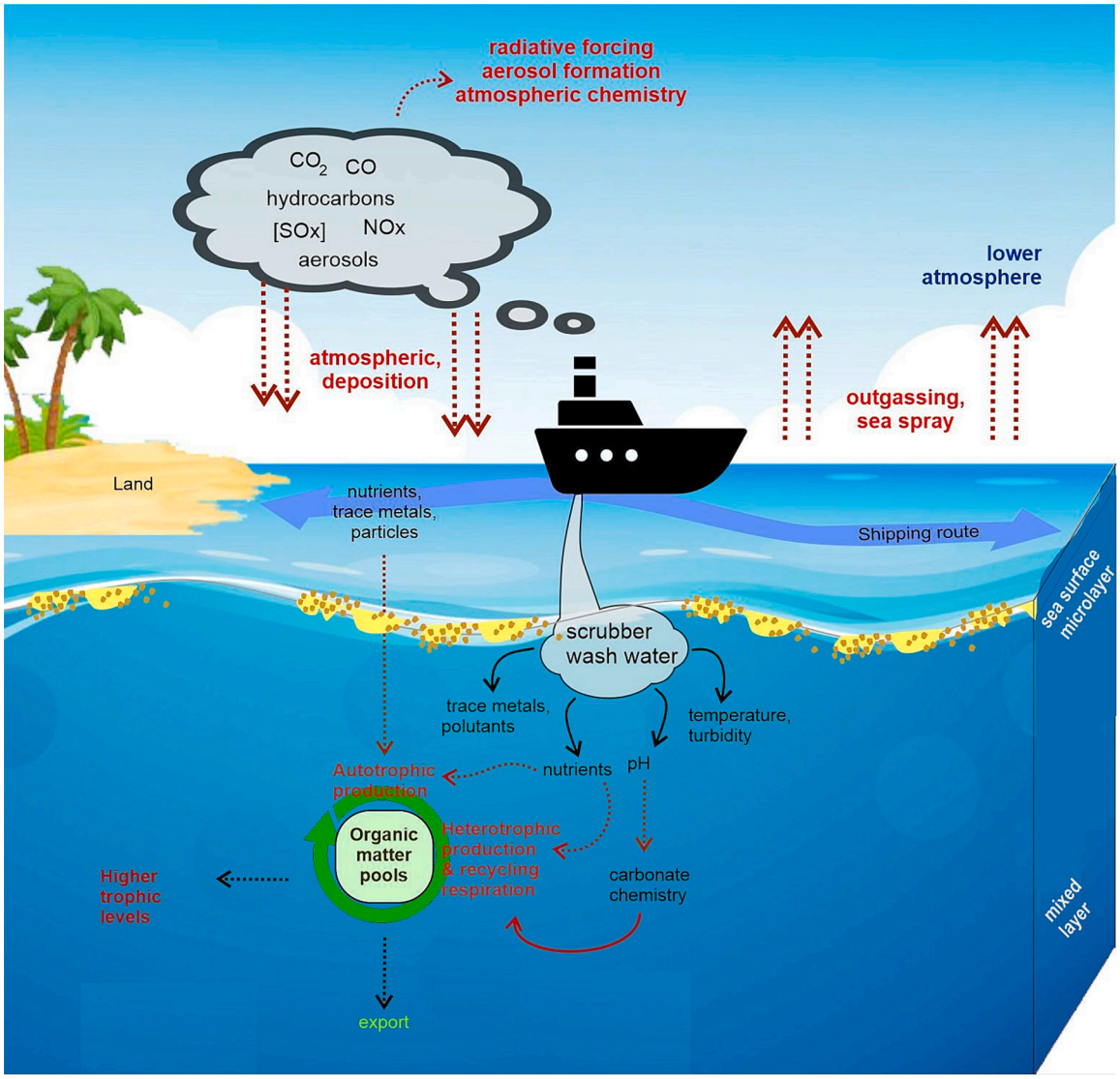


Fig. 3. Detailed mechanism of how marine transportation affects the environment, modified from [36].

from respiratory causes [27].

4. Possible substitutes for conventional fuels used in the shipping industry

Conventional marine fuels, such as heavy fuel oil, marine diesel oil, and marine gasoline oil, have been used to power ships for more than a century [65]. These conventional types of marine fuel can be purchased virtually anywhere in the world. However, in comparison to the fuels used in other industries, such as transportation or aviation, these fuels have typically undergone a lower level of refining or processing. For instance, heavy fuel oil has a very high viscosity and contains considerable amounts of sulphur, both of which, when burned, create toxic emissions of sulphur oxide [72]. Because of rules imposed by the maritime industry on sulphur oxide and nitrogen oxide emissions from ships, certain types of non-conventional fuels, such as low sulphur heavy fuel oil, liquefied natural gas, and methanol have been utilised as a substitute for conventional marine fuels throughout the course of the past two decades [38].

Emerging alternative fuels such as biodiesel, hydrogenated vegetable oil, Fischer-Tropsch (F-T) diesel, hydrogen, and ammonia are likely to be the primary pathways for achieving future low carbon or zero carbon shipping over the next three decades and through this, assist in meeting the emissions reduction targets set by the IMO for 2030 and 2050[8]. Biodiesel, hydrogenated vegetable oil and F-T diesel can all be derived from several types of feedstock. The comparative characteristics of conventional, non-conventional and prospective alternative fuel is presented in Table 1. Even though the vast majority of alternative fuels are not produced using resources originating from fossil fuels, some of these fuels may nevertheless produce upstream emissions during their production. Not every-one of these possibilities has attained its full potential yet in the market. The distinctive feature of different potential alternative fuels is shown in Fig. 5 [5,56].

4.1. Biofuels

Biofuels can provide significant reductions in greenhouse gas emissions, and in some instances, they can be utilised as a “drop-in” fuel, meaning that current engines require very little modification for these fuels to function [48]. Conventional first-generation biofuels such as straight vegetable oil, hydrotreated vegetable oil, biodiesel, and bioethanol, are widely available [69,70]. However, these biofuels cannot be used on a global scale because of sustainability concerns. Many of the problems with first-generation biofuels can be mitigated by second-generation biofuels, but these fuels are still confronted with technological and financial obstacles [14,52]. Waste oil can be used to help

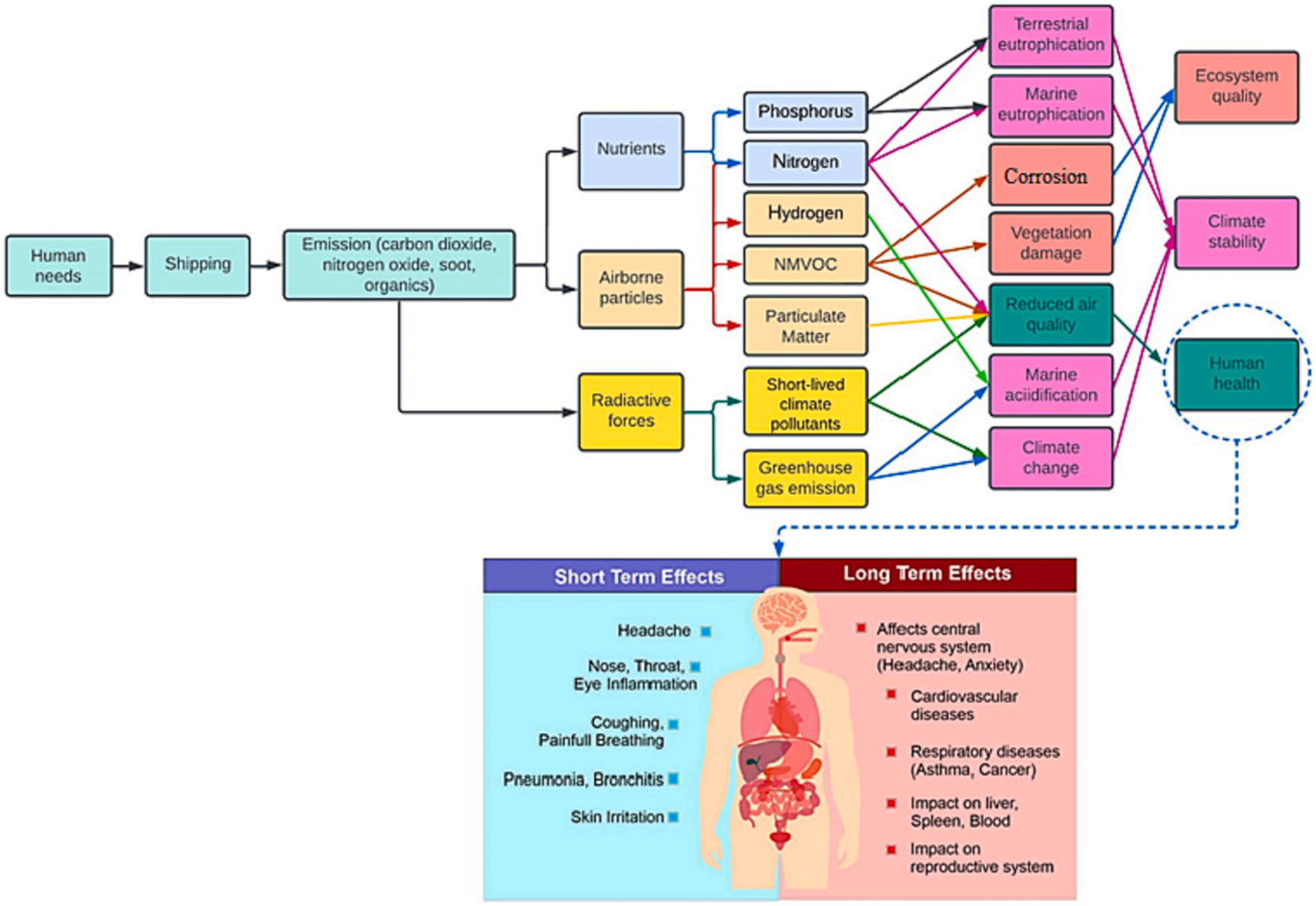


Fig. 4. Effect of harmful emissions such as non-methane volatile organic compounds (NMVOC) on environmental and human health, modified from [3,99].

alleviate these problems, but there is not enough of it to meet the needs of mass production. Advanced biofuels are produced from feedstocks that pose fewer environmental risks. Fischer-Tropsch diesel, pyrolysis oil, and lignocellulosic ethanol are the most promising advanced biofuels for use in global shipping.

The maritime industry has only in recent years began to experiment with various kinds of biofuels by evaluating engine compatibility, examining fuel properties, and constructing real-world pilot projects. The majority of the demonstration projects carried out thus far include mixing first-generation biodiesel with conventional bunker fuel. Fuel suppliers have started selling biofuels that may be used directly as a replacement for heavy fuel oil, and they are attempting to cut the price of these biofuels so that they can be utilised more widely. Additionally, producers of engines are focusing on the development of new engines able to operate on various alternative fuels without modification.

In comparison to first- and second-generation biofuels, advanced biofuels produce fewer greenhouse gas emissions. Using biofuels could also aid in the fight against nitrogen oxides and sulphur oxides. Sulphur is extremely uncommon in biofuels [42]and in comparison to marine gas oil, biodiesel emits fewer nitrogen oxides and particulate matter [94]. Biodiesel also has a very low sulphur concentration (20 ppm) [42].

4.2. Natural gas

Natural gas (liquefied or compressed) is commonly used as a maritime fuel. When compared to other hydrocarbon fuels, natural gas has the highest energy density as measured by mass and the lowest carbon emissions for each unit of energy release [11]. It may be possible to produce natural gas by converting fossil fuels or biomass. The long-term potential of natural gas derived from fossil sources is somewhat restricted in the shipping industry; thus, natural gas from biomass sources is preferred. Natural gas has the potential to reduce the carbon footprint of shipping by up to 25 % compared to current marine fuel oils [41]. However, actual implementation is expected to yield a decrease of 12–20 % due to the risk of methane slip, which has a global warming potential of 25 [53,89].

In recent years, liquefied natural gas has become increasingly popular as a maritime fuel. When compressed into liquid form, the volume required is only 1/600th of the gaseous product, making it ideal for use as a bunker on ships [93]. The benefit of adopting LNG stems more from environmental considerations than from financial ones. The price of liquefied natural gas is far more affordable in comparison to the price of feedstock for many other alternative fuels. In addition to this, liquefied natural gas should be stored in insulated tanks to eliminate the possibility of leaks [56]. The use of liquefied natural gas as ship fuel has the potential to greatly reduce emissions of nitrogen oxides, particulate matter, and greenhouse gas emissions [97]. When switching from heavy fuel oil to liquefied natural gas, emissions of nitrogen oxide, sulphur oxide, particulate matter, and carbon dioxide are reduced by 85–90 %, 100 %, and 15–20 %, respectively [93].

4.3. Hydrogen

Hydrogen has the simplest chemical structure of all elements and is the most abundant element on earth [46]). It also has the highest energy storage capacity of all the fuels compared to its weight. However, it occurs most frequently in the form of compounds, which necessitates an energy demand to extract it. The raw materials used in the creation of

Table 1
The comparative characteristics of different alternative fuels for the shipping industry [5,32,71,96].

Potential alternative fuels	CF	D (kg/m^3)	CN	BP (°C)	AIT (°C)	F (vol)%		Emissions intensity while using in marine engine			
						Min.	Max.	CO_2	SOx	NOx	PM
Ammonia	NH_3	0.73	120*	−33	651	15.0	28.0				
Biodiesel	C_{16}–C_{18}	880**	50**	>180	261	0.6	7.5				
Ethanol	C_2H_5OH	789	10**	78	365	3.3	19.0				
F-T diesel	C_{15}–C_{18}	778**	77**	>180	204	0.6	7.5				
Hydrogen	H_2	0.09	>130*	−253	585	4.0	75.0				
Hydrotreated vegetable oil	C_{15}–C_{18}	780**	>70	>180	204	0.6	7.5				
Low sulphur heavy fuel oil	C_8–C_{25}	993**	>20	>180	230	0.6	7.5				
Marine diesel oil	C_{10}–C_{15}	819**	>35	>180	210	0.6	7.5				
Methanol	CH_3OH	792	<5	65	464	6.7	36.0				
Natural gas	CH_4	0.78	130*	−162	540	5.0	15.0				
Straight vegetable oil	C_{14}–C_{22}	930**	38**	>180	424	0.6	7.5				

CF = Chemical formula; D = Density; CN = Cetane number; BP = Boiling point; AIT = Autoignition temperature (in air); F = Flammability (in air); red label = high; orange label = medium; green label = low; *= Octane number; **= Average value.

hydrogen are various types of fossil fuels, such as natural gas, coal, biomass, and water [2].Only water and heat are produced by the oxidation process when hydrogen is burned as fuel; no other emissions are produced [45]. The natural occurrence of hydrogen in its elemental state is uncommon on earth, despite the fact that hydrogen is the most common chemical element in the universe. Nevertheless, hydrogen is present in a wide variety of hydrocarbons, as well as water and manufactured compounds. Hydrogen's high energy density, which ranges from 120 to 142 MJ/kg [21], is one of the most significant advantages it offers. This high energy density, coupled with the development of mature production techniques, positions hydrogen favourably for use both as a seasonal energy storage option and as an alternative fuel [74].

A recent assessment by Bicer and Dincer [16] found that the production of hydrogen using hydropower can result in ten times less carbon dioxide emissions than those produced by heavy fuel oil across the whole heavy fuel oil life cycle. In addition, the use of hydrogen in

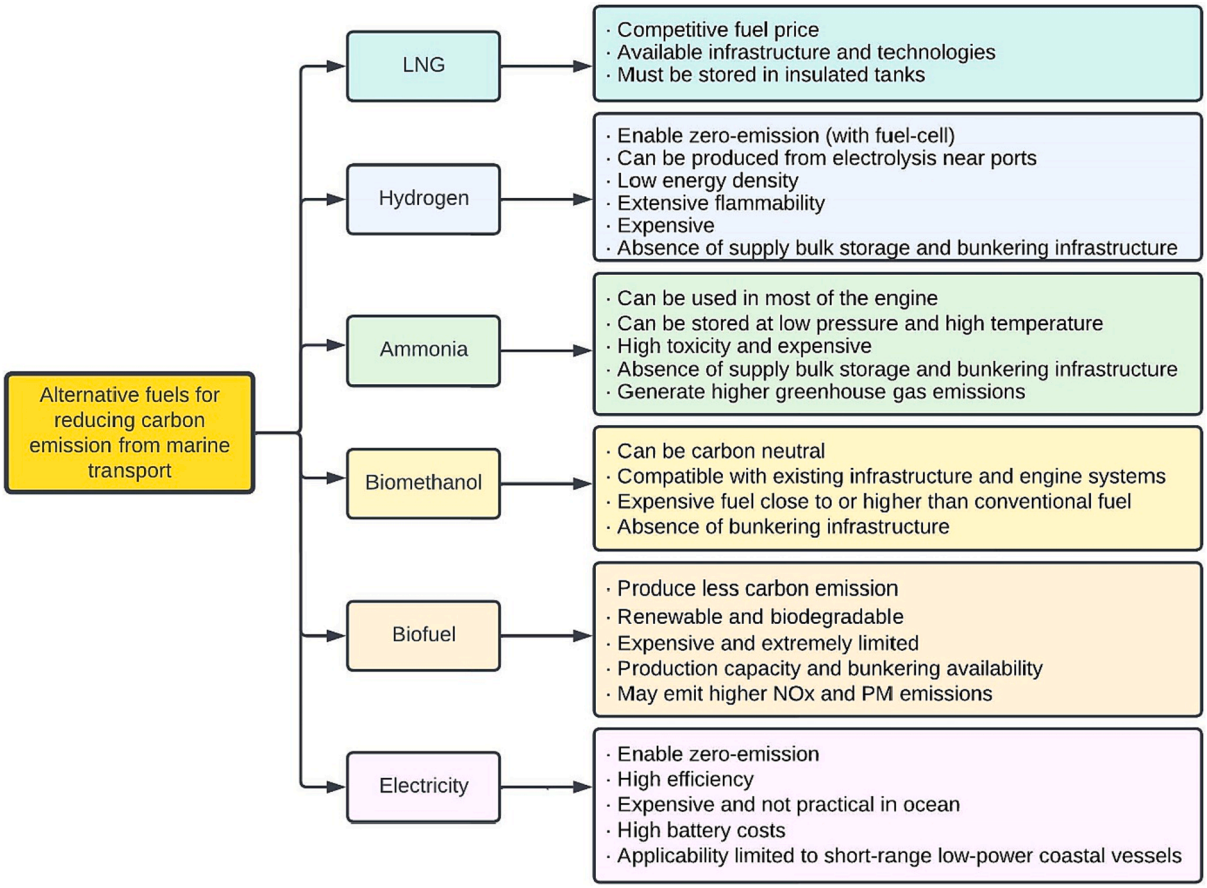


Fig. 5. The distinctive feature of different potential alternative fuels for the shipping industry [5,56].

conjunction with heavy fuel oil (which accounts for 50 % of the total fuel) has the potential to cut carbon dioxide emissions by as much as 43 % every tonne-kilometre. This demonstrates that it is possible to achieve a considerable decrease in carbon dioxide emissions and other greenhouse gas emissions even when only a portion of the present conventional marine fuels are replaced with hydrogen.

4.4. Ammonia

Ammonia is a chemical substance that is made up of hydrogen and nitrogen, but no carbon. Ammonia is primarily employed in the manufacturing of fertilisers in addition to a wide variety of other industrial applications, including plastics, synthetic fibres, and explosives [95]. Ammonia can be found in nature, but its production for commercial purposes requires high temperatures, a lot of pressure, and the catalytic reactions of nitrogen and hydrogen. In the year 2020, the actual global output of ammonia was 187 Mt, and of that amount, 85 % was used either directly or indirectly in agriculture as a variety of chemical fertilisers [102].

Recent years have seen a rise in interest in the use of ammonia as an alternative fuel due to the favourable features it possesses as a clean energy carrier. Ammonia does not contain any carbon, and as a result, when it is burned, only water and nitrogen are produced as by-products; this makes it a highly environmentally benign fuel. When used as a fuel, ammonia does not release carbon dioxide, sulphur oxide, or particulate matter into the atmosphere. The production of ammonia using heavy fuel oil or coal would result in higher levels of carbon dioxide emissions for each unit of energy produced. Electrolysis processes, on the other hand, have the potential to produce ammonia from renewable energy. Because the transition from tank to propeller does not result in the emission of any carbon, the later pathway, which is a carbon-free one, would make it possible to produce ammonia without creating any carbon emissions. Bicer and Dincer [16] discovered that the use of ammonia as a dual fuel together with heavy fuel oil can result in a 27 % reduction of carbon dioxide emissions per tonne-kilometre over the course of the whole fuel life cycle. However, the utilisation of ammonia produced by wind energy as a dual fuel, where the proportion of ammonia to hydrofluorooctane is split 50/50, has the potential to cut overall carbon dioxide emissions from the fuel life cycle by up to 34.5 % per tonne-kilometre.

4.5. Methanol

Methanol is a liquid fuel that has a low flashpoint, does not contain any sulphur, and is easy to store. It also produces less emissions and has a smaller carbon footprint compared to conventional marine fuels. Methanol can be used as ship fuel, and there is at least one marine engine on the market that has the potential to use methanol as one of its dual fuels [96]. According to the findings, methanol is the “best alternative fuel” because of its “rapid availability,” “use of existing infrastructure”, “cheap”, and “simplicity of engine design and ship technology” [40]. To date, 11 ships that run on methanol have been put into service [11]. In comparison to heavy fuel oil or marine gasoline oil, the burning of methanol in marine engines results in only a slight decrease in carbon dioxide emissions but a low level of emissions of other contaminants [44]. It is believed that the Stena Germanica, the world’s first methanol-powered sea vessel, has cut its emission of sulphur oxide by 99 %, nitrogen oxide by 60 %, particulate by 95 %, and carbon dioxide by 25 %, thereby complying with the most recent emission control areas standards on its journey through the Baltic Sea [19]. As another example, the Effship project, assessed various technical solutions and marine fuels able to meet the sulphur oxide and nitrogen oxide reductions rules “in

the short term" (2015–2016), as well as greenhouse gas reduction targets "in the medium term" (2030), and "in the long term" (2050).

Methanol can be created from a wide variety of sources, such as natural gas, via the catalytic hydrogenation of a waste carbon dioxide stream, or even from biomass [96]. When using biomass as the feedstock, the produced carbon dioxide emissions are considered biogenic and can be deducted. In the present day, that the yearly production of methanol is projected as>100 million tonnes; however, the majority of that production is not renewable and is obtained from natural gas and coal. It is anticipated that the costs associated with producing methanol from renewable sources are anywhere from 1.5 to 4 times greater than the cost associated with producing methanol from fossil fuels [6]. Methanol could be made available in major port terminals across the world, because the methanol production operates on a worldwide scale. It would be worthwhile to research the prospective uses of methanol as a maritime fuel in the short-term, medium-term, and long-term as methanol is in steady supply and is widely distributed.

4.6. Wind power

Wind-assisted ship propulsion (WASP) technology could potentially expedite the shipping industry's attempts to decarbonize by using the wind to replace part of the propulsive power currently generated by fossil fuels used in ships. This could help accelerate the decarbonization process. According to Chou et al.[26], a decrease in potential emissions can be accomplished with the use of WASP technologies, which have the potential to reduce carbon dioxide emissions by>20 %. This finding is consistent with past research which has highlighted WASP technology as a viable alternative for improving the energy efficiency profile of the maritime transport industry by reducing the amount of carbon dioxide emissions produced by the industry's operations. As a result, WASP technology has a place in the transition away from carbon-based fuels, as well as in the reduction of greenhouse gas emissions produced by ships for each unit of transport work carried out in accordance with the IMO's greenhouse gas emission reduction strategy. Additionally, it has the potential to serve as a valuable complement to the use of alternative fuels. The proposal that the Comoro Islands made to the recently held MEPC 75 of the IMO [39]put the enticing role that wind may play in the process of decarbonization officially on the table where policy decisions are made [29].

There are currently-two vendors whose products are on the verge of being commercially viable for commercial ships. Further, a wide variety of technologies currently being tested should be commercially viable by the year 2035. Auxiliary towing kites manufactured by the German company SkySails have been installed on two multi-purpose ships and one bulk carrier. Cylinders for rotors have also been put through their paces by the shipping company Bore, which installed on several cargo ship that travelled between Rotterdam and Teesport in the United Kingdom [60].

4.7. Solar energy

It is possible for photovoltaic energy to supply enough electricity to meet the on-board power demand of a ship. Solar panels have the ability to be mounted on deck in either a horizontal or vertical orientation, depending on the type of sail being used. This technique can also be used in conjunction with wind technology, which consists of a structure similar to a mast, to expand the solar energy collection area. The efficiency of this technique is likewise dependent on prevailing weather conditions, just like wind energy is. Solar energy output reaches higher values in regions close to the equator, which makes this a more viable investment for ships that operate in these regions. The potential savings from solar energy are not particularly extensive. Smith et al.[84]make the assumption that there will be a reduction in auxiliary engine fuel usage of between 0.1 % and 3 %. Bauman et al. [20] reported that the possible carbon dioxide reduction that has been reported in a variety of research ranges from 0.2 to 12 %. The operators of the Japanese car carrier Nichioh Maru, which runs with solar aid, state that the technology enables a reduction of 1,400 tonnes of diesel per year, which translates to a yearly reduction of 4,200 tonnes of carbon dioxide emissions.

5. UN sustainable development goal (SDG) realization

Alternative energy has the potential to reduce harmful emissions by up to 100 % depending on the sources and it could therefore align well with the ambitions of SDG-13 (climate action) and several targets categorized within different goals, for example: SDG-7 (ensure access to affordable, reliable, sustainable and modern energy for all) Target 7.2 (by 2030, increase substantially the share of renewable energy in the global energy mix) and Target 7.3 (by 2030, double the global rate of improvement in energy efficiency); SDG-3 (good health and well-being) Target 3.9 (by 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination); SDG-6 (clean water and sanitation) Target 6.3 (by 2030, improve water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally) and Target 6.6 (by 2020, protect and restore water-related ecosystems, including mountains, forests, wetlands, rivers, aquifers and lakes); SDG-11 (sustainable cities and communities) Target 11.6 (by 2030, reduce the adverse per capita environmental impact of cities, including by paying special attention to air quality and municipal and other waste management).

6. Key challenges and recommendations for alternative energy sources to decarbonize the shipping industry

While alternative fuels have the potential to cut the greenhouse gas emissions of the shipping industry, they face a number of problems that must be addressed before they can reach their full potential. The following are the specific issues associated with alternate energy sources in shipping industry. The challenges of alternative fuels reducing greenhouse gas emission in shipping industry are summarized in Table 2.

- There is a possibility that the available supply of biofuels will not be adequate to power the entire global marine fleet. The existing supply of biofuels, which includes both biodiesel and bioethanol, can only satisfy roughly 15 % of total demand [98]. This estimate is based on what can be produced by adopting current agricultural and forestry production technology without increasing land consumption or decreasing food availability. More optimistic estimates, such as those presented in [1]indicate that there will be enough biofuels to meet future demand, whereas more conservative estimates, such as those presented by [84], indicate that biofuels will only be able to power a small portion of the global marine fleet.

However, advanced biofuels present an option that reduces emissions and has the potential to become more broadly available if the necessary investment is made. Advanced biofuels are currently being used on a number of ships, despite the fact that further research on their functionality and physical characteristics, as well as additional testing and standardisation may be necessary [5]. Numerous regions all over the world are currently engaged in the manufacturing of biofuel on a massive scale [22].

The use of biodiesel in marine engines is not hindered by significant technical or financial hurdles. Nonetheless, only biodiesel derived from waste is anticipated to generate significant life-cycle GHG reductions in comparison to distillate fuel. Furthermore, in order for biodiesel to be used in existing marine engines, it must be blended with conventional marine fuels up to a particular ratio. However, the potential for biodiesel

Table 2
Potential and limitations of alternative fuels in the shipping industry in cutting GHG emissions.

Alternative energy sources	Potential reduction of GHG emissions		Reported challenges	Refs.
	Min.	Max.		
Solar power	0.20	12	There are insufficient batteries to store enough electricity to travel a long distance	[5,20]
Fuel cells	2	20	Fuel cell capacities, price, and operational life	[5,20]
Wind power	1	50	Controlling larger sails is difficult due to the instability they introduce, the amount of crew work required, and lack of control over wind direction	[5,20]
Biofuel including biomethanol, biodiesel, SVO	25	100	Oxidation, environmental implications, availability of feedstock, technical and operational concerns, and economic factors	[34,57]
Liquefied natural gas	5	30	Political, infrastructure, and transportation cost	[20,57,63]
Hydrogen	1	43	Expensive storage technology e and storage safety issues	[16,57,81]
Ammonia	27	34.70	Storage at cryogenic temperatures, bunkering, the cost of infrastructure, corrosiveness, leaking, toxicity, and explosiveness	[17,50,57]
Electricity	0	100	Costs associated with high power charging requirements	[57]

to reduce emissions throughout the entire sector is diminished due to this constraint.

- The calorific value of liquefied natural gas is 33.75 MJ/m^3, significantly lower than the calorific value of conventional oil at 38.44 MJ/L. This indicates that the former utilises a greater quantity of fuel in order to accomplish the same level of power output as the latter [92]. The claim that low liquefied natural gas prices are the key impetus to push the fuel into the maritime fuel mix has therefore been debunked by this evidence. Additionally, substantial initial investments are necessary for high-tech liquefied natural gas engines and cryogenic double-walled fuel tanks [93]. Liquefied natural gas, on the other hand, is the most widely used alternative fuel to marine bunker oil. There are already 165 ships in operation and 154 confirmed orders for liquefied natural gas and its uses can be found all over the world and in the majority of ship categories [75]. The most significant drawback of liquefied natural gas is that it is not capable of meeting the requirements of the Initial Strategy of the IMO, which calls for a reduction of 50 %in carbon dioxide emissions.
- There is still a need for research and development to make hydrogen a practical alternative to traditional diesel fuel, particularly in terms of commercial scale feasibility. There is currently no standardised design or fuelling technique for hydrogen-powered ships or the associated bunkering infrastructure [66].

The volumetric energy density of hydrogen has a significant impact on both its transportation and storage. As a result of its low volumetric energy density, a greater volume is required for both transport and storage. As a result, hydrogen in both compressed and liquid forms is required for use in practical applications [61]. Despite this, the amount of storage space necessary for compressed hydrogen is anywhere from ten to twenty times greater than that required for marine fuel oils, depending on the pressure (250–700 bar). Due to the high pressure and extremely low storage temperature, there are stringent standards that must be met by both transport vehicles and fuel tanks [77]. This results in significant operational costs. Hydrogen's economic competitiveness is further impeded by the fact that hydrogen fuel has less space for payload and requires more frequent refuelling.

In addition, solutions for the unresolved safety design challenges pertaining to the volatile nature of the fuel are needed. That being said, a number of nations like Japan, China, India, and Australia are investing in research and development to manufacture hydrogen in a sustainable manner [4]. Over the course of the next three years, the Government of the Republic of India has allocated a total of 2.5 billion rupees, which is equivalent to $58 million. Another nation that has shown how national initiatives effectively initiate a shift to hydrogen fuels is Australia. As a result of the efforts Australia has put into collaborating with regional and global businesses, it is anticipated that it will emerge as the market leader in this sector.

- The majority of ammonia uses fossil fuels as its source of production. Studies on international shipping have evaluated the prospect of utilising hydrogen mixed with ammonia as a potential means to reduce carbon dioxide emissions by 70 % by the year 2035 [49,51]. A scenario in which carbon neutrality is maintained for Danish marine freight until 2050 was constructed by Hansson et al.[50]. However, they did mention a few things that need to be taken into consideration, such as the safe transportation of fuel and the research and development of fuel cells. According to Yusuf and Ibrahim [17]the use of ammonia in dual-fuel marine engines could result in a reduction of up to 33.5 % of the total global GHG emissions (using geothermal-based ammonia). On the other hand, significant nitrogen oxide emissions are a cause for concern with this fuel type. In addition, the findings of the study indicated that the utilisation of hydrogen as well as ammonia in maritime transportation could have a sizeable impact on greenhouse gas emission levels.
- Methanol has a flash point that is lower than average, it is poisonous, and the vapour it produces is thicker than air [31]. Since the energy density of methanol is around half that of standard marine fuels, nearly twice as much capacity is required for its storage compared to marine fuels. Even though it is easy to store methanol, there is potential for corrosion since methanol is incompatible with certain materials [34]. In addition, methanol is a substance that is acutely toxic to people. As a result, essential protective measures and contingency plans are required for potential spill events to eliminate the possibility of inhalation, ingestion, and skin contact with the substance.
- The most significant downside of wind energy is its intermittent nature which makes it challenging to extract the maximum amount of electricity from it when sailing [62]. Both primary and secondary propulsion can be accomplished with the help of wind. If existing ships are going to make use of wind power for primary and auxiliary propulsion, the ship's structural stability needs to be taken into consideration. It is also possible that retrofitting certain technologies, such as Flettner rotors, onto existing ships is not possible in every circumstance.
- Solar cells have two fundamental problems: (1) there is not enough space on a ship to install the cells; and (2), the ships do not have enough batteries to store an adequate amount of energy [73]. Intermittency problems and the possibility of solar panels being corroded as a result of the presence of salt in the water are two additional challenges to overcome [59]. It is possible to say that photovoltaic (PV) systems are generally the most promising technology making use of renewable energy due to recent developments in the field (for instance, the continual improvement in efficiency) and the rapid growth of the systems. As a result of the corrosive nature of salty water, photovoltaic systems installed on ships require

additional research and development on the materials used. As a result, the potential of PV systems installed on ships is lower than terrestrial PV systems.

7. Conclusions

The environmental problems caused by the shipping industry are a primary source of worry in the modern era. The IMO is acting to promote and support the industry during the transition towards major reductions in greenhouse gas emissions. Since it is anticipated that the internal combustion engine will continue to play an important role in the medium term, the shipping industry will need to transition away from its reliance on fossil fuels and towards the use of alternative fuels to fulfil these goals. In spite of these goals, there has not yet been any considerable shift towards the use of alternative fuels on a commercial basis. This study thus reviewed the potential benefits of using alternative energy sources in the shipping industry for lowering carbon dioxide emissions, as well as the potential challenges that may arise.

Liquefied natural gas is the alternative fuel for ships that has received the most attention. The primary issue with the liquefied natural gas is that it cannot fulfil the IMO's Initial Strategy, which demands a 50 % decrease in carbon dioxide emissions. Scientists are increasingly focusing their attention on methanol, ammonia, and hydrogen fuels. Biofuels like biomethanol and biodiesel can help drastically reduce greenhouse gas emissions in the shipping industry by as much as 100 %. However, it is not recommended to use biofuels in the maritime industry because of their limited supply and low competitiveness in comparison to other energy sources. In order to reduce the amount of emissions produced by the shipping industry from an environmental, technological, and economic standpoint, there has been an increased focus on the feasibility of substituting alternative fuels for traditional marine fuels in recent years. As a guide towards advancing the development of low-carbon shipping, the findings of this research help to identify the areas with the most potential for emission reduction.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Further reading

[30] [Record #939 is using a reference type undefined in this output style].